

School of Computer Science, UPES Dehradun

Research Methodology in Computer Science

Assignment: Unit I & II Tasks

Dr. Ayush Kumar Agrawal

Instructions

- Students must work in groups of 3–4.
- Each group must **choose one research topic of their interest** (e.g., Artificial Intelligence, IoT, Cybersecurity, Robotics, Algorithms, Software Engineering, Cloud Computing).
- All tasks must be performed **on the selected topic**.
- Deliverables: A PowerPoint Presentation (7–10 slides) + a short written summary (1 page).

Unit I: Research Problem Formulation

Task 1: Problem Identification

- (a) Frame three possible **research problem statements** in your chosen topic.
- (b) Define the **objectives** of one selected problem.
- (c) Highlight potential **errors in problem formulation**.

Task 2: SMART Framework

Convert your selected topic into a research problem using the **SMART** framework (Specific, Measurable, Achievable, Relevant, Time-bound). Present the refined problem statement in 3–4 PPT slides.

Task 3: Peer Critique Activity

Within your group, review each other's problem statements and critique them for:

- Clarity
- Scope
- Feasibility

- Relevance

Summarize your critique and final refined problem statement in PPT.

Unit II: Research Methods in CS

Task 4: Comparing Proofs

Take one claim related to your chosen topic and:

- Provide a **theoretical or mathematical justification**.
- Provide an **empirical justification** (if possible, simulate/collect data or cite existing results).
- Conclude: Which proof is more convincing for your case? Why?

Task 5: Dialectical Reasoning

Identify a controversial research question in your chosen topic.

- Present arguments **in favor**.
- Present arguments **against**.
- Arrive at a **refined conclusion**.

Task 6: Mini Research Project

On your selected topic:

- Clearly state the research problem.
- Apply at least two research approaches (e.g., empirical + theoretical OR deduction + induction).
- Present the process and results in PPT (7–10 slides).

Evaluation Rubric (Total: 25 Marks)

Criteria	Marks
Clarity and Precision of Research Problem	5
Application of Research Methods (Unit II concepts)	5
Depth of Analysis and Critical Thinking	5
Quality of PPT (organization, visuals, structure)	5
Originality, Teamwork, and Presentation Skills	5
Total	25 Marks

Submission Guidelines

- Submit PPT and summary report in PDF format.
- Mention group members' names and roll numbers on the first slide.
- Deadline: To be announced in class.